29/01/2024
Morning

Corporate Office : Aakash Tower, 8, Pusa Road, New Delhi-110005 | Ph.: 011-47623456

Answers & Solutions

Time : 3 hrs. *for* M.M. : 300

JEE (Main)-2024 (Online) Phase-1

(Mathematics, Physics and Chemistry)

IMPORTANT INSTRUCTIONS:

(1) The test is of **3 hours** duration.

(2) This test paper consists of 90 questions. Each subject (MPC) has 30 questions. The maximum marks are 300.

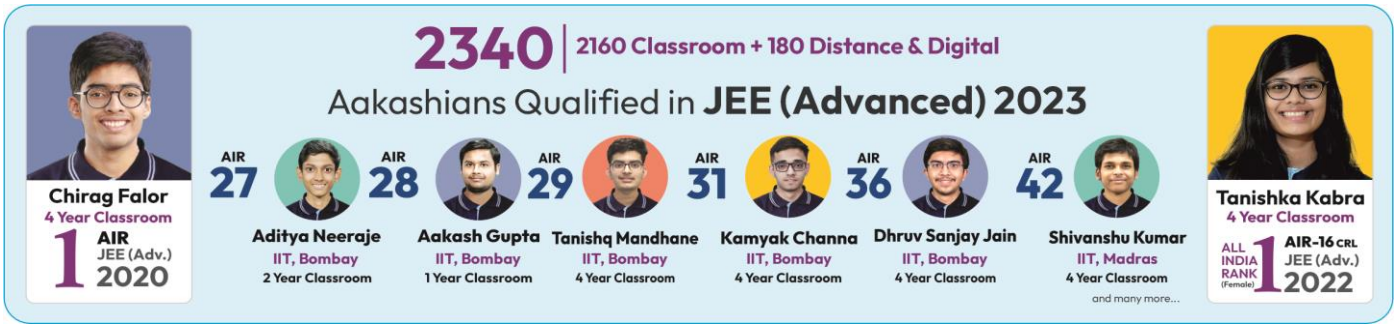
(3) This question paper contains **Three Parts**. **Part-A** is Mathematics, **Part-B** is Physics and **Part-C** is Chemistry. Each part has only two sections: **Section-A** and **Section-B**.

(4) **Section - A :** Attempt all questions.

(5) **Section - B :** Attempt any 05 questions out of 10 Questions.

(6) **Section - A (01 – 20)** contains 20 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

(7) **Section - B (21 – 30) contains** 10 **Numerical value** based questions. The answer to each question should be rounded off to the **nearest integer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.



MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. If the value of the integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{x^2\cos x}{1+\pi^x}+\frac{1+\sin^2 x}{1+e^{\sin x^{2023}}}\right)dx=\frac{\pi}{4}(\pi+a)-2$, then the value of a is

(1) $\frac{3}{2}$ (2) 3

(3) 2 (4) $-\frac{3}{2}$

Answer (2)

Sol. $I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{x^2\cos x}{1+\pi^x}+\frac{1+\sin^2 x}{1+e^{\sin x^{2023}}}\right)dx$

$I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{x^2\cos x}{1+\pi^{-x}}+\frac{1+\sin^2 x}{1+e^{-\sin x^{2023}}}\right)dx$

$2I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(x^2\cos x+\sin^2 x+1\right)dx$

$2I=2\int_{0}^{\frac{\pi}{2}}\left(x^2\cos x+\sin^2 x+1\right)dx$

$I=\int_{0}^{\frac{\pi}{2}}\left(x^2\cos x+\frac{1-\cos 2x}{2}+1\right)dx$

$=\int_{0}^{\frac{\pi}{2}}\left(x^2\cos x+\frac{3}{2}-\frac{\cos 2x}{2}\right)dx$

$=x^2\sin x-\int 2x\sin x\,dx+\frac{3}{2}x-\frac{\sin 2x}{4}$

$=x^2\sin x-\left[2x(-\cos x)-\int 2(-\cos x)dx+\frac{3x}{2}-\frac{\sin 2x}{4}\right]$

$=\left[x^2\sin x+2x\cos x-2\sin x+\frac{3x}{2}-\frac{\sin 2x}{4}\right]_0^{\frac{\pi}{2}}$

$=\frac{\pi^2}{4}+\frac{3\pi}{4}-2$

$\Rightarrow a=3$

2. For $x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, if $y(x)=\int\frac{\operatorname{cosec}x+\sin x}{\operatorname{cosec}x\sec x+\tan x\sin^2 x}dx$, and $\lim\limits_{x\to\left(\frac{\pi}{2}\right)^-}y(x)=0$ then $y\left(\frac{\pi}{4}\right)$ is equal to

(1) $\frac{1}{2}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (2) $-\frac{1}{2}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(3) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (4) $\frac{1}{\sqrt{2}}\tan^{-1}\left(-\frac{1}{2}\right)$

Answer (4)

Sol. $y(x)=\int\frac{\operatorname{cosec}x+\sin x}{\operatorname{cosec}x\sec x+\tan x\sin^2 x}dx$

$=\int\frac{\frac{1}{\sin x}+\sin x}{\frac{1}{\sin x\cos x}+\frac{\sin^3 x}{\cos x}}dx$

$=\int\frac{\frac{1+\sin^2 x}{\sin x}}{\frac{1+\sin^4 x}{\sin x\cos x}}dx=\int\frac{(1+\sin^2 x)\cos x}{1+\sin^4 x}dx$



Put $\sin x = t$

$\therefore \quad \cos x\, dx = dt$

$$\int \frac{1+t^2}{1+t^4}dt = \int \frac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}}dt$$

Let $t - \frac{1}{t} = u$

$$\left(1+\frac{1}{t^2}\right)dt = du$$

and $\left(t-\frac{1}{t}\right)^2 = u^2 \Rightarrow t^2+\frac{1}{t^2} = u^2+2$

$$\therefore \quad \int \frac{du}{u^2+2} = \frac{1}{\sqrt{2}}\tan^{-1}\frac{4}{\sqrt{2}} + c$$

$$y(x) = \frac{1}{\sqrt{2}}\tan^{-1}\frac{\left(\sin x - \frac{1}{\sin x}\right)}{\sqrt{2}} + c$$

$$\lim_{x\to\left(\frac{\pi}{2}\right)^-} \frac{1}{\sqrt{2}}\tan^{-1}\frac{\left(\sin x - \frac{1}{\sin x}\right)}{\sqrt{2}} + c = 0$$

$$\lim_{x\to\left(\frac{\pi}{2}\right)^-} \frac{1}{\sqrt{2}}\tan^{-1}\frac{\sin^2 x - 1}{\sqrt{2}\sin x} + c = 0$$

$\Rightarrow \quad c = 0$

$$\therefore \quad y(x) = \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\sin^2 x - 1}{\sqrt{2}\sin x}\right)$$

$$y\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\frac{1}{2}-1}{\sqrt{2}\times\frac{1}{\sqrt{2}}}\right)$$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{-1}{2}\right)$$

3. In a ΔABC, suppose $y = x$ is the equation of the bisector of the angle B and the equation of the side AC is $2x - y = 2$. If $2AB = BC$ and the points A and B are respectively (4, 6) and (α, β), then $\alpha + 2\beta$ is equal to

(1) 39

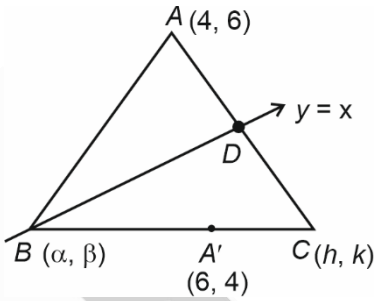
(2) 42

(3) 48

(4) 45

Answer (2)

Sol.



$\because$ B lies on $y = x$

$\therefore \quad \alpha = \beta$ ...(1)

Let coordinate of C be (h, k)

Here $AB : CB = 1 : 2 = AD : CD$

Equation of line $AC : 2x - y = 2$ ...(1)

and $BD : x = y$ ...(2)

From (1) and (2) coordinate of $D = (2, 2)$

$\because \quad \left(\frac{h+8}{3}, \frac{k+12}{3}\right) = (2, 2)$

$\therefore \quad (h, k) = (-2, -6)$

Equation of line $BA'C$ is : $y - 4 = \frac{5}{4}(x-6)$

$\because \quad x = y = \alpha \Rightarrow \alpha = 14$

$\therefore \quad \alpha + 2\beta = 3 \times 14 = 42$



4. $\lim\limits_{x\to\frac{\pi}{2}}\left(\dfrac{1}{\left(x-\frac{\pi}{2}\right)^2}\int\limits_{x^3}^{\left(\frac{\pi}{2}\right)^3}\cos\left(t^{\frac{1}{3}}\right)dt\right)$ is equal to

(1) $\dfrac{3\pi}{4}$

(2) $\dfrac{3\pi}{8}$

(3) $\dfrac{3\pi^2}{4}$

(4) $\dfrac{3\pi^2}{8}$

Answer (4)

Sol. $\lim\limits_{x\to\frac{\pi}{2}}\left(\dfrac{1}{\left(x-\frac{\pi}{2}\right)^2}\int\limits_{x^3}^{\left(\frac{\pi}{2}\right)^3}\cos\left(t^{\frac{1}{3}}\right)dt\right)$

Applying L-H rule

$$\lim_{x\to\frac{\pi}{2}}\frac{\cos\left(\frac{\pi}{2}\right)\times 0-\cos x\left(3x^2\right)}{2\left(x-\frac{\pi}{2}\right)(1)}$$ (by Leibnitz theorem)

$$=\lim_{x\to\frac{\pi}{2}}\left(-\frac{\sin\left(\frac{\pi}{2}-x\right)}{\left(x-\frac{\pi}{2}\right)}\right)\left(\frac{3}{2}x^2\right)$$

$$=\frac{3}{2}\left(\frac{\pi}{2}\right)^2$$

$$=\frac{3\pi^2}{8}$$

So, option (4) is correct.

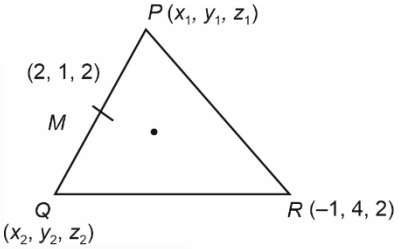
5. Let PQR be a triangle with $R(-1, 4, 2)$. Suppose $M(2, 1, 2)$ is the mid-point of PQ. The distance of the centroid of ΔPQR from the point of intersection of the lines $\dfrac{x-2}{0}=\dfrac{y}{2}=\dfrac{z+3}{-1}$ and $\dfrac{x-1}{1}=\dfrac{y+3}{-3}=\dfrac{z+1}{1}$ is

(1) $\sqrt{69}$ (2) 69

(3) $\sqrt{99}$ (4) 9

Answer (1)

Sol.



Given two lines are

$$\frac{x-2}{0}=\frac{y}{2}=\frac{z+3}{-1}=\lambda(\text{let})$$

$$\frac{x-1}{1}=\frac{y+3}{-3}=\frac{z+1}{1}=\mu(\text{let})$$

for point of intersection

$x = 2, y = 2\lambda, z = -\lambda - 3$ (General point of line 1)

$x = \mu + 1, y = -3\mu - 3, z = \mu - 1$ (General point of line 2)

$2=\mu+1\Rightarrow\boxed{\mu=1}$

$2\lambda=-3\mu-3\Rightarrow\boxed{\lambda=-3}$

Point of intersection of two lines is $(2, -6, 0)$

$\because$ M is the mid point of PQ,

$\therefore$ $x_1 + x_2 = 4, y_1 + y_2 = 2, z_1 + z_2 = 4$

Centroid of ΔPQR



$= \left(\frac{x_1+x_2-1}{3}, \frac{y_1+y_2+4}{3}, \frac{z_1+z_2+2}{3}\right)$

$= (1, 2, 2)$

$\therefore$ distance between (1, 2, 2) and (2, –6, 0)

$= \sqrt{1^2+8^2+2^2} = \sqrt{69}$

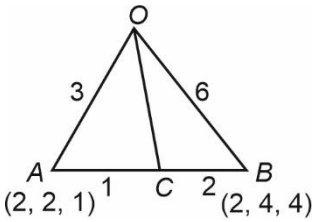
6. Let O be the origin and the position vectors of A and B be $2\hat{i}+2\hat{j}+\hat{k}$ and $2\hat{i}+4\hat{j}+4\hat{k}$ respectively. If the internal bisector of $\angle AOB$ meets the line AB at C, then the length of OC is

(1) $\frac{2}{3}\sqrt{34}$ (2) $\frac{3}{2}\sqrt{34}$

(3) $\frac{3}{2}\sqrt{31}$ (4) $\frac{2}{3}\sqrt{31}$

Answer (1)

Sol.



Position vector of $C = \frac{6\hat{i}+8\hat{j}+6\hat{k}}{3}$

$\therefore \quad |\overrightarrow{OC}| = \frac{1}{3}\sqrt{36+64+36}$

$= \frac{1}{3}\sqrt{136}$

$= \frac{2}{3}\sqrt{34}$

7. A function $y = f(x)$ satisfies $f(x)\sin 2x + \sin x - (1 + \cos^2 x) f'(x) = 0$ with condition $f(0) = 0$. Then, $f\left(\frac{\pi}{2}\right)$ is equal to

(1) –1 (2) 1

(3) 2 (4) 0

Answer (1)

Sol. $y \cdot \sin 2x + \sin x - (1+\cos^2 x)\frac{dy}{dx} = 0$

$\left(1+\cos^2 x\right)\frac{dy}{dx} - (\sin 2x)y = \sin x$

$\frac{dy}{dx} - \left(\frac{\sin 2x}{1+\cos^2 x}\right)y = \frac{\sin x}{1+\cos^2 x}$

It is of the form : $\frac{dy}{dx} + P(x)y = Q(x)$

I.F = $e^{\int P(x)dx} = e^{-\int \frac{\sin 2x}{1+\cos^2 x}dx}$

Let $I = \int \frac{\sin 2x}{1+\cos^2 x}dx = 2\int \frac{\sin x \cos x}{1+\cos^2 x}dx$

$= -\int \frac{2tdt}{1+t^2} \quad \begin{bmatrix} \cos x = t \\ -\sin x dx = dt \end{bmatrix}$

$= -\int \frac{du}{u} \quad \begin{bmatrix} 1+t^2 = u \\ 2tdt = du \end{bmatrix}$

$= -\ln u$

$= -\ln|1 + t^2|$

$I = -\ln|1 + \cos^2 x|$

$\therefore \quad \text{I.F} = e^{\ln|1+\cos^2 x|} = 1+\cos^2 x$

$\Rightarrow \quad y \cdot (\text{I.F}) = \int Q(x)(\text{I.F})dx$

$y(1+\cos^2 x) = \int \frac{\sin x}{(1+\cos^2 x)}(1+\cos^2 x)dx$

$y(1 + \cos^2 x) = -\cos x + c$

$\because \quad f(0) = 0; \quad c = 1$

$\Rightarrow \quad y(1 + \cos^2 x) = -\cos x + 1$

Now, $f\left(\frac{\pi}{2}\right)$ is

$\Rightarrow \quad y\left(1+\cos^2\frac{\pi}{2}\right) = -\cos\frac{\pi}{2} + 1$

$\Rightarrow \quad y = 1$



8. If $z=\frac{1}{2}-2i$ is such that $|z+1| = \alpha z + \beta(1+i)$, $i=\sqrt{-1}$ and $\alpha, \beta \in \mathbb{R}$, then $\alpha+\beta$ is equal to

(1) –4 (2) 2

(3) –1 (4) 3

Answer (4)

Sol. $\left|\frac{1}{2}-2i+1\right| = \alpha\left(\frac{1}{2}-2i\right)+\beta(1+i)$

$\sqrt{\frac{9}{4}+4} = \alpha\left(\frac{1}{2}-2i\right)+\beta(1+i)$

$\frac{5}{2} = \alpha\left(\frac{1}{2}\right)+\beta+i(-2\alpha+\beta)$

$\frac{\alpha}{2}+\beta=\frac{5}{2}$ …(i)

$-2\alpha+\beta=0$ …(ii)

Solving (i) and (ii)

$\frac{\alpha}{2}+2\alpha=\frac{5}{2}$

$\frac{5}{2}\alpha=\frac{5}{2}$

$\alpha = 1$ and $\beta = 2$

$\therefore \quad \alpha+\beta=3$

9. Consider the function $f:\left[\frac{1}{2},1\right]\to\mathbb{R}$ defined by $f(x)=4\sqrt{2}x^3-3\sqrt{2}x-1$. Consider the statements

(I) The curve $y = f(x)$ intersects the x-axis exactly at one point.

(II) The curve $y = f(x)$ intersects the x-axis at $x=\cos\frac{\pi}{12}$.

Then

(1) Both (I) and (II) are incorrect.

(2) Only (II) is correct.

(3) Only (I) is correct.

(4) Both (I) and (II) are correct.

Answer (4)

Sol. Let $x=\frac{\pi}{12}$

$\therefore \quad 3x=\frac{\pi}{4}$

$\cos(3x)=\frac{1}{\sqrt{2}}$

$\Rightarrow 4\cos^3 x-3\cos x=\frac{1}{\sqrt{2}}$

$\Rightarrow 4\sqrt{2}\cos^3 x-3\sqrt{2}\cos x-1=0$

$\therefore \quad x=\left(\frac{\pi}{12}\right)$ is the solution of above equation

statement II is correct

$f'(x)=12\sqrt{2}x^2-3\sqrt{2}$

Which is positive for $x\in\left[\frac{1}{2},1\right]$

Hence, only one solution in $\left[\frac{1}{2},1\right]$

$\therefore$ Statement I is also correct

10. In an A.P., the sixth term $a_6 = 2$. If the product $a_1a_4a_5$ is the greatest, then the common difference of the A.P. is equal to

(1) $\frac{5}{8}$ (2) $\frac{2}{3}$

(3) $\frac{3}{2}$ (4) $\frac{8}{5}$

Answer (4)

Sol. Let first term = a

Common difference = d

Given, $a+5d=2$ …(1)

Product $(P) = a_1a_5a_6 = a(a+4d)(a+3d)$



From (1)

$P = (2 - 5d)(2 - d)(2 - 2d)$

$$\frac{dP}{dd} = (2-5d)(2-d)(-2) + (2-5d)(2-2d)(-1)$$
$$+ (-5)(2-d)(2-2d)$$

$= -2[(d-2)(5d-2) + (d-1)(5d-2) + 5(d-1)(d-2)]$

$= -2[5d^2 + 4 - 12d + 5d^2 + 2 - 7d + 5d^2 + 10 - 15d]$

$= -2[15d^2 - 34d + 16]$

$\Rightarrow d = \frac{8}{5}$ or $\frac{2}{3}$

at $\left(\frac{8}{5}\right)$, Product attains maxima

Answer is $\frac{8}{5}$

11. If $\alpha, -\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ is the solution of $4\cos\theta + 5\sin\theta = 1$, then the value of $\tan\alpha$ is

(1) $\frac{10-\sqrt{10}}{6}$ (2) $\frac{\sqrt{10}-10}{12}$

(3) $\frac{\sqrt{10}-10}{6}$ (4) $\frac{10-\sqrt{10}}{12}$

Answer (2)

Sol. $\alpha, -\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ is solution of $4\cos\theta + 5\sin\theta = 1$

So, $4\cos\alpha + 5\sin\alpha = 1$

Divide both sides by $\cos\alpha$

$$\left(\text{as } \alpha \neq \frac{\pi}{2}, -\frac{\pi}{2}, \text{so we can divide by } \cos\alpha\right)$$

$\Rightarrow 4 + 5\tan\alpha = \sec\alpha$

$\Rightarrow 4 + 5\tan\alpha = \sqrt{1+\tan^2\alpha}$

Square both sides,

$\Rightarrow 16 + 25\tan^2\alpha + 40\tan\alpha = 1 + \tan^2\alpha$

$\Rightarrow 24\tan^2\alpha + 40\tan\alpha + 15 = 0$

$\Rightarrow \tan\alpha = \frac{-40 \pm \sqrt{1600 - 4 \times 24 \times 15}}{48}$

$\Rightarrow \tan\alpha = \frac{-10 \pm \sqrt{10}}{12}$

So $\tan\alpha = \frac{-10+\sqrt{10}}{12}$

12. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \alpha & \beta \\ 0 & \beta & \alpha \end{bmatrix}$ and $|2A|^3 = 2^{21}$ where $\alpha, \beta \in \mathbb{Z}$, then a value of α is

(1) 5 (2) 9

(3) 17 (4) 3

Answer (1)

Sol. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \alpha & \beta \\ 0 & \beta & \alpha \end{bmatrix}$

$2A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2\alpha & 2\beta \\ 0 & 2\beta & 2\alpha \end{bmatrix}$

$|2A| = 2(4\alpha^2 - 4\beta^2)$

$|2A|^3 = 8(4\alpha^2 - 4\beta^2)^3 = 2^{21}$ (given)

$\Rightarrow (4\alpha^2 - 4\beta^2)^3 = 2^{18}\,(2^6)^3$

$\Rightarrow 4(\alpha^2 - \beta^2) = 2^6$

$\Rightarrow \alpha^2 - \beta^2 = 2^4$

$\Rightarrow \alpha^2 - \beta^2 = 16$

Now as $\alpha, \beta \in Z$

$(\alpha, \beta) = (5, 3)$ is a possible pair.

So $\alpha = 5$



13. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-zero vectors such that $\vec{b}$ and $\vec{c}$ are non-collinear. If $\vec{a}+5\vec{b}$ is collinear with $\vec{c}, \vec{b}+6\vec{c}$ is collinear with $\vec{a}$ and $\vec{a}+\alpha\vec{b}+\beta\vec{c}=\vec{0}$, then $\alpha+\beta$ is equal to

(1) 35 (2) –30

(3) –25 (4) 30

Answer (1)

Sol. $\vec{a}+5\vec{b}$ is collinear with $\vec{c}$

$\vec{a}+5\vec{b}=\lambda\vec{c}$

$\vec{a}=\lambda\vec{c}-5\vec{b}$ ...(i)

$\vec{b}+6\vec{c}$ is collinear with $\vec{a}$

$\vec{b}+6\vec{c}=\mu\vec{a}$

$\vec{b}+6\vec{c}=\mu(\lambda\vec{c}-5\vec{b})$

$\vec{b}+6\vec{c}=\mu\lambda\vec{c}-5\mu\vec{b}$

$\vec{b}(1+5\mu)=\vec{c}(\lambda\mu-6)$

$\Rightarrow$ $1+5\mu=0$

$\Rightarrow$ $\mu=-\frac{1}{5}$ and $\lambda\mu-6=0$

$\lambda\left(-\frac{1}{5}\right)=6$

$\lambda=-30$

By equation (i), we get :

$\vec{a}+5\vec{b}+30\vec{c}=0$

Compare it with

$\vec{a}+\alpha\vec{b}+\beta\vec{c}=0$

$\alpha=5, \beta=30$

$\Rightarrow$ $\alpha+\beta=35$

14. If $f(x)=\begin{cases}2+2x, & -1\le x<0\\ 1-\frac{x}{3}, & 0\le x\le 3\end{cases}$;

$g(x)=\begin{cases}-x, & -3\le x\le 0\\ x, & 0<x\le 1\end{cases}$, then range of $(fog)(x)$ is

(1) [0, 1] (2) (0, 1]

(3) [0, 1) (4) [0, 3)

Answer (1)

Sol. $f(x)=\begin{cases}2+2x, & -1\le x<0\\ 1-x/3 & 0\le x\le 3\end{cases}$

$g(x)=\begin{cases}-x, & -3\le x\le 0\\ x & 0<x\le 1\end{cases}$ $\Rightarrow$ $g(x)$ is nothing but a part of $|x|$

$\Rightarrow$ $g(x) = |x|, x \in [-3, 1] \Rightarrow g(x) \ge 0$ $(\because |x| \ge 0)$

$\Rightarrow$ $f(g(x))=\begin{cases}2+2|x|, & -1<|x|<0\\ 1-\frac{|x|}{3}, & 0\le|x|\le 3\end{cases}$

$\Rightarrow$ $f(g(x))=1-\frac{|x|}{3}, |x|\in[0,3]$

Range of $f(g(x))$

$|x| \in [0, 3] \Rightarrow -\frac{[x]}{3} \in [-1, 0]$

$1-\frac{[x]}{3}\in[0,1]$

Range of $f(g(x))$ = [0, 1]

15. If in a G.P. of 64 terms, the sum of all the terms is 7 times the sum of the odd terms of the G.P. then the common ratio of the G.P. is equal to

(1) 6 (2) 7

(3) 4 (4) 5

Answer (1)

Sol. Let the G.P. is $a, ar, ar^2, ...$



As given

$$\frac{a(r^{64}-1)}{r-1} = 7.\frac{a((r^2)^{32}-1)}{r^2-1}$$

$\Rightarrow r^2 - 1 = 7r - 7$

$\Rightarrow r^2 - 7r + 6 = 0$

$\Rightarrow r = 1$ (Rejected) or $r = 6$.

16. Suppose $f(x) = \frac{(2^x + 2^{-x})\tan x\sqrt{\tan^{-1}(x^2 - x + 1)}}{(7x^2 + 3x + 1)^3}$.

Then the value of $f'(0)$ is equal to

(1) $\frac{\pi}{2}$ (2) $\sqrt{\pi}$

(3) 0 (4) π

Answer (2)

Sol. Let $f(x) = \tan x\, g(x)$

where $g(x) = \frac{(2^x + 2^{-x})\sqrt{\tan^{-1}(x^2 - x + 1)}}{(7x^2 + 3x + 1)^3}$

$\Rightarrow f'(x) = \sec^2 x g(x) + \tan x g'(x)$

$f'(0) = \sec^2(0)g(0) + \tan 0\, g'(0)$

$= g(0)$

$$g(0) = \frac{(2^0 + 2^{-0})\sqrt{\tan^{-1}(0^2 - 0 + 1)}}{(7(0)^2 + 3(0) + 1)^3}$$

$$= 2\sqrt{\tan^{-1}1} = 2\sqrt{\frac{\pi}{4}} = \sqrt{\pi}$$

17. Let A be a square matrix such that $AA^T = I$. Then $\frac{1}{2}A\left[(A + A^T)^2 + (A - A^T)^2\right]$ is equal to

(1) $A^2 + A^T$ (2) $A^2 + I$

(3) $A^3 + I$ (4) $A^3 + A^T$

Answer (4)

Sol. Let $B = \frac{1}{2}A\left[(A + A^T)^2 + (A - A^T)^2\right]$

$$B = \frac{1}{2}A(A^2 + (A^T)^2 + AA^T + A^TA + A^2 + (A^T)^2 - AA^T - A^TA)$$

$$B = \frac{1}{2}A(2A^2 + 2(A^T)^2) = A(A^2 + A^T \cdot A^T)$$

$= A^3 + AA^TA^T = A^3 + A^T$

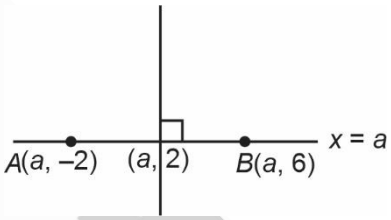
18. Let $\left(5, \frac{a}{4}\right)$ be the circumcentre of a triangle with vertices $A(a, -2)$, $B(a, 6)$ and $C\left(\frac{a}{4}, -2\right)$. Let α denote the circumradius, β denote the area and γ denote the perimeter of the triangle. Then $\alpha + \beta + \gamma$ is

(1) 62 (2) 60

(3) 30 (4) 53

Answer (4)

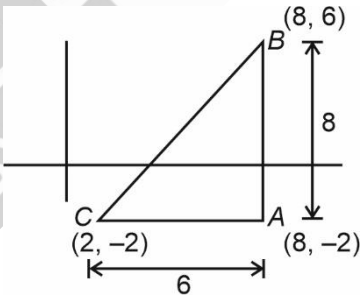
Sol.



Circumcentre lie on perpendicular bisector of $AB \Rightarrow y = 2$

$\Rightarrow \frac{a}{4} = 2 \Rightarrow a = 8$

$\Rightarrow$



$\Rightarrow$ Perimeter = 6 + 8 + 10 = 24

Area = $\frac{6 \times 8}{2} = 24$

Circumradius = half of hypotenuse

$= \frac{10}{2} = 5$

$\alpha + \beta + \gamma \Rightarrow 24 + 24 + 5 = 53$



19. Let R be a relation on $\mathbb{Z} \times \mathbb{Z}$ defined by $(a, b)\ R\ (c, d)$ if and only if $ad - bc$ is divisible by 5. Then R is

(1) Reflexive and symmetric but not transitive

(2) Reflexive but neither symmetric nor transitive

(3) Reflexive and transitive but not symmetric

(4) Reflexive, symmetric and transitive

Answer (1)

Sol. $(a, b)\ R\ (c, d) \Rightarrow ad - bc$ is divisible by 5

$= cb - da$ is also divisible by 5

$= (c, d)\ R\ (a, b)$

$\Rightarrow R$ is symmetric relation

If $(a, b)\ R\ (a, b) \Rightarrow ab - ba = 0$ is divisible by 5

$\Rightarrow R$ is reflexive

If (a, b) R $(c, d) \Rightarrow (ad - bc)$ is divisible by 5

and $(c, d)\ R\ (e, f) \Rightarrow (cf - de)$ is divisible by 5

$\Rightarrow (a, b)\ R\ (e, f) = (af - be)$ needs to be multiple of 5

$ad - bc = 5k_1$

$cf - de = 5k_2$

$afd - bcf = 5fk_1$

$bcf - bde = 5bk_2$

$\Rightarrow d(af - be) = 5(fk_1 + bk_2)$

But if $d = 0$ then there can be $af - be$ such that it is not divisible by 5

$\Rightarrow$ Not transitive

For example take : $(a, b) \equiv (6, 1)$, $(c, d) = (0, 0)$, $(e, f) = (1, 7)$

20. A fair die is thrown until 2 appears. Then the probability, that 2 appears in even number of throws, is

(1) $\frac{6}{11}$ (2) $\frac{5}{6}$

(3) $\frac{5}{11}$ (4) $\frac{1}{6}$

Answer (3)

Sol. A : 2 appears in even number of throws

B : 2 appears on die

$$\Rightarrow P(B) = \frac{1}{6},\quad P(\bar{B}) = \frac{5}{6}$$

$$P(A) = P(\bar{B})P(B) + P(\bar{B})P(\bar{B})P(\bar{B})P(B) +$$

$$P(\bar{B})P(\bar{B})P(\bar{B})P(\bar{B})P(\bar{B})P(B) + \ldots.$$

$$\Rightarrow P(A) = \frac{5}{6}\cdot\frac{1}{6} + \left(\frac{5}{6}\right)^3\cdot\frac{1}{6} + \left(\frac{5}{6}\right)^5\frac{1}{6} + \ldots$$

$$\frac{\frac{5}{36}}{1-\frac{5^2}{6^2}} = \frac{\frac{5}{36}}{\frac{36-25}{36}} = \frac{5}{11}$$

SECTION - B

Numerical Value Type Questions: This section contains 10 Numerical based questions. Attempt any 5 questions out of 10. The answer to each question should be rounded-off to the nearest integer.

21. If $\frac{{}^{11}C_1}{2} + \frac{{}^{11}C_2}{3} + \ldots + \frac{{}^{11}C_9}{10} = \frac{n}{m}$ with $gcd(n, m) = 1$, then $n + m$ is equal to _____.

Answer (2041)

Sol. $(1+x)^{11} = {}^{11}C_0 + {}^{11}C_1x + {}^{11}C_2x^2 + \ldots + {}^{11}C_{11}x^{11}$

$$\int_0^1 (1+x)^{11}\,dx = {}^{11}C_0x + \frac{{}^{11}C_1x^2}{2} + \frac{{}^{11}C_2x^3}{3} + \ldots + \frac{{}^{11}C_9x^{10}}{10} + \frac{{}^{11}C_{10}x^{11}}{11} + \frac{{}^{11}C_{11}x^{12}}{12}\Bigg]_0^1$$

$$\therefore \frac{{}^{11}C_1}{2} + \frac{{}^{11}C_2}{3} + \ldots + \frac{{}^{11}C_9}{10} = \frac{2^{12}-1}{12} - 1 - 1 - \frac{1}{12}$$

$$= \frac{2035}{6} = \frac{n}{m}$$

$n + m = 2041$

22. Let α, β be the roots of the equation $x^2 - x + 2 = 0$ with $\text{Im}(\alpha) > \text{Im}(\beta)$. Then $\alpha^6 + \alpha^4 + \beta^4 - 5\alpha^2$ is equal to _____.

Answer (13)

Sol. $x^2 + 2 = x$

$\Rightarrow x^4 + 4 + 4x^2 = x^2$



So,

$x^4 = -4 - 3x^2$

$\Rightarrow x^6 = -4x^2 - 3x^4$

$\Rightarrow x^6 = 12 + 5x^2$

Now,

$\alpha^6 - 5\alpha^2 = 12$

So $\alpha^4 + \beta^4 + 12$

$= 4 - 3(\alpha^2 + \beta^2)$

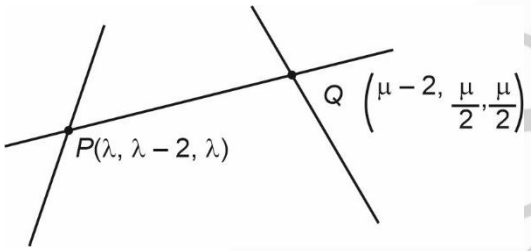
$= 4 - 3[1 - 4]$

$= 13$

23. A line with direction ratios 2, 1, 2 meets the lines $x = y + 2 = z$ and $x + 2 = 2y = 2z$ respectively at the points P and Q. If the length of the perpendicular from the point (1, 2, 12) to the line PQ is l, then l^2 is _____.

Answer (65)

Sol.



$x = y + 2 = z = \lambda$

$x + 2 = 2y = 2z = \mu$

Now PQ is proportional to <2, 1, 2>

$$\therefore \frac{\lambda-\mu+2}{2} = \frac{\lambda-2-\frac{\mu}{2}}{1} = \frac{\lambda-\frac{\mu}{2}}{2}$$

$\therefore \mu = 4, \lambda = 6$

$\therefore P(6, 4, 6)$ and $Q(2, 2, 2)$

$$L_{PQ} = \frac{x-2}{2} = \frac{y-2}{1} = \frac{z-2}{2} = K$$

Point of PQ : $(2K + 2, K + 2, 2K + 2)$

$< 2K + 1, K, 2K - 10 >$

$2(2K + 1) + K + 2(2K - 10) = 0$

$9K - 18 = 0 \Rightarrow \boxed{K = 2}$

$\therefore$ Point on $PQ \equiv (6, 4, 6)$

$l^2 = (6 - 1)^2 + (4 - 2)^2 + (12 - 6)^2$

$= 25 + 4 + 36 = 65$

24. All the letters of the word "GTWENTY" are written in all possible ways with or without meaning and these words are written as in a dictionary. The serial number of word "GTWENTY" is _____.

Answer (553)

Sol. GTWENTY

(1) $E : \frac{6!}{2!} = 360$

(2) $\overline{GE} : \frac{5!}{2!}, \overline{GN} : \frac{5!}{2!}$

(3) $GTE : 4!$, $GTN : 4!$, $GTT : 4!$

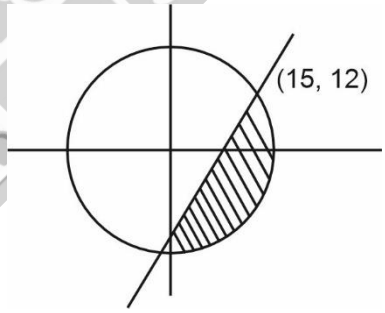
(4) GTWENTY = 1

$\Rightarrow 360 + 60 + 60 + 24 + 24 + 24 + 1 = 553$

25. The area (in sq. units) of the part of the circle $x^2 + y^2 = 169$ which is below the line $5x - y = 13$ is $\frac{\pi\alpha}{2\beta} - \frac{65}{2} + \frac{\alpha}{\beta}\sin^{-1}\left(\frac{12}{13}\right)$, where α, β are coprime numbers. Then $\alpha + \beta$ is equal to _____.

Answer (171)

Sol.



$$\text{Area} = \frac{\pi(13)^2}{2} - \left[\frac{1}{2}\times 25\times 5 + \int_{12}^{13}\sqrt{169-y^2}\,dy\right]$$

$$\frac{169\pi}{4} - \frac{65}{2} + \frac{169}{2}\sin^{-1}\frac{12}{13}$$



$\Rightarrow \alpha = 169$

$\beta = 2$

$\therefore \alpha + \beta = 171$

26. If the mean and variance of the data 65, 68, 58, 44, 48, 45, 60, α, β, 60 where $\alpha > \beta$ are 56 and 66.2 respectively, then $\alpha^2 + \beta^2$ is equal to _____.

Answer (6344)

Sol. $56 = \frac{65+68+58+44+48+45+60+60+\alpha+\beta}{10}$

$\Rightarrow \alpha + \beta = 112$

$66.2 \times 10 = (65-56)^2 + (68-56)^2 + (58-56)^2 + (44-56)^2 + (48-56)^2 + (45-56)^2 + (60-56)^2 + (60-56)^2 + (\alpha-56)^2 + (\beta-56)^2$

$662 = 81 + 144 + 4 + 144 + 64 + 121 + 16 + 16 + 2 \times (56)^2 + \alpha^2 + \beta^2 - 112(\alpha+\beta)$

$\Rightarrow \alpha^2 + \beta^2 = 6344$

27. If the solution curve $y = y(x)$ of the differential equation $(1 + y^2)(1 + \log_e x)\,dx + x\,dy = 0$, $x > 0$ passes through the point (1, 1) and $y(e) = \frac{\alpha - \tan\left(\frac{3}{2}\right)}{\beta + \left(\frac{3}{2}\right)}$ then $\alpha + 2\beta$ is

Answer (3)

Sol. $(1 + y^2)(1 + \ln x)\,dx + x\,dy = 0$

Put $1 + \ln x = t \Rightarrow \frac{1}{x}dx = dt$

$(1 + y^2)\,t\,dt + dy = 0$

$t\,dt = -\frac{dy}{1+y^2}$

$\frac{t^2}{2} = -\tan^{-1}(y) + C$

$\Rightarrow \frac{(1+\ln x)^2}{2} = -\tan^{-1} y + C$ passes through (1, 1)

$\Rightarrow C = \frac{1}{2} + \frac{\pi}{4}$

$\Rightarrow (1 + \ln x)^2 = -2\tan^{-1} y + 1 + \frac{\pi}{2}$

Put $x = e$

$\Rightarrow 4 = -2\tan^{-1}(y(e)) + 1 + \frac{\pi}{2}$

$\tan^{-1}(y(e)) = \frac{\pi}{4} - \frac{3}{2}$

$y(e) = \frac{1 - \tan\left(\frac{3}{2}\right)}{1 + \tan\left(\frac{3}{2}\right)}$

$\Rightarrow \alpha + 2\beta = 3$

28. Equations of two diameters of a circle are $2x - 3y = 5$ and $3x - 4y = 7$. The line joining the points $\left(-\frac{22}{7}, -4\right)$ and $\left(-\frac{1}{7}, 3\right)$ intersects the circle at only one point $P(\alpha, \beta)$. Then $17\beta - \alpha$ is equal to _____.

Answer (2)

Sol. Intersection point of diameters $2x - 3y = 5$ and $3x - 4y = 7$ is centre of circle $C(1, -1)$

Equation of line joining $\left(-\frac{22}{7}, 4\right)$ and $\left(-\frac{1}{7}, 3\right)$ is

$y = \frac{7}{3}x + \frac{10}{3} \Rightarrow 7x - 3y + 10 = 0$

Radius of circle is $\frac{20}{\sqrt{58}}$

$\Rightarrow$ Equation of circle is $(x-1)^2 + (y+1)^2 = \frac{400}{58}$



$$x^2 + y^2 - 2x + 2y - \frac{142}{29} = 0$$

$\Rightarrow$ Tangent at $P(\alpha, \beta)$ is

$$\alpha x + \beta y - (\alpha + x) + (\beta + y) - \frac{142}{29} = 0$$

$$\Rightarrow \quad \frac{\alpha - 1}{7} = \frac{\beta + 1}{-3} = \frac{-\alpha + \beta - \frac{142}{29}}{10}$$

$$\Rightarrow \quad \alpha = -\frac{41}{29}, \beta = \frac{1}{29}$$

$$\Rightarrow \quad 17\beta - \alpha = \frac{17}{29} + \frac{41}{29} = 2$$

29. If the points of intersection of two distinct conics $x^2 + y^2 = 4b$ and $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ lie on the curve $y^2 = 3x^2$ then $3\sqrt{3}$ times the area of the rectangle formed by the intersection points is _____.

Answer (432)

Sol. $x^2 + y^2 = 4b$ ...(i)

$$\frac{x^2}{16} + \frac{y^2}{b^2} = 1, \quad \text{...(ii)}$$

$y^2 = 3x^2$ ...(iii)

From (i) and (iii)

$x^2 = b$, $y^2 = 3b$

From (ii)

$$\frac{b}{16} + \frac{3b}{b^2} = 1$$

$$\frac{b}{16} + \frac{3}{b} = 1$$

$b^2 - 16b + 48 = 0$

$b = 12, b = 4$

(i) & (ii) are distinct $\Rightarrow b \neq 4$, $b = 12$

$\Rightarrow x = \pm 2\sqrt{3}$, $y = \pm 6$

$\Rightarrow 3\sqrt{3}\ A = 3\sqrt{3} \times 4\sqrt{3} \times 12 = 432$

30. Let $f(x) = 2^x - x^2$, $x \in R$. If m and n are respectively the number of points at which the curves $y = f(x)$ and $y = f'(x)$ intersect the x-axis then the value of $m + n$ is _____

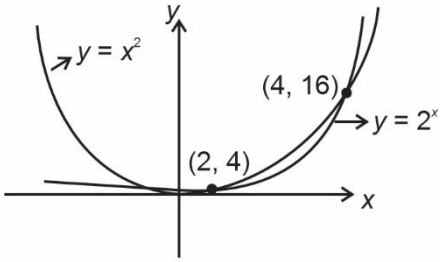
Answer (5)

Sol. $f(x) = 2^x - x^2$

$f'(x) = 2^x \ln 2 - 2x$

When $f(x) = 0$

$2^x = x^2$

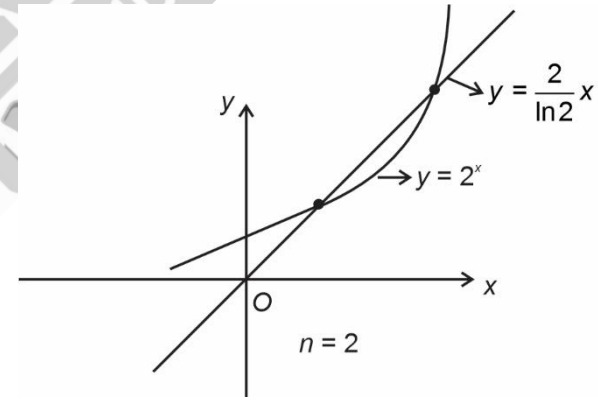


$\Rightarrow m = 3$

When $f'(x) = 0$

$2^x \ln 2 - 2x = 0$

$$2^x = \frac{2}{\ln 2} x$$



$\Rightarrow m + n = 5$



PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

31. The deflection in moving coil galvanometer falls from 25 divisions to 5 division when a shunt of 24 Ω is applied. The resistance of galvanometer coil will be :

(1) 100 Ω (2) 12 Ω

(3) 96 Ω (4) 48 Ω

Answer (3)

Sol. $i \propto$ number of divisions

$\therefore \quad 5(R_G) = (25 - 5)\,(24)$

$R_G = 96\ \Omega$

32. Match List-I with List-II

	List-I		List-II
a.	$\oint \vec{B}\cdot\vec{dl} = \mu_0 i_c + \mu_0\varepsilon_0 \frac{d\phi_E}{dt}$	(i)	Gauss' law for electricity
b.	$\oint \vec{E}\cdot\vec{dl} = \frac{d\phi_B}{dt}$	(ii)	Gauss' law for magnetism
c.	$\oint \vec{E}\cdot\vec{dA} = \frac{Q}{\varepsilon_0}$	(iii)	Faraday law
d.	$\oint B\cdot\vec{dA} = 0$	(iv)	Ampere - Maxwell las

Choose the correct answer from the options given below :

(1) a(iv), b(iii), c(i), d(ii) (2) a(ii), b(iii), c(i), d(iv)

(3) a(iv), b(i), c(iii), d(ii) (4) a(i), b(ii), c(iii), d(iv)

Answer (1)

Sol. $\oint \vec{E}\cdot\vec{dx} = \frac{Q}{\varepsilon_0}$ (Gauss's law)

Similarly,

$\oint \vec{E}\cdot\vec{dl} = \frac{d\phi_B}{dt}$ (Faraday law)

$\therefore$ a(iv), b(iii), c(i), d(ii)

33. At what distance above and below the surface of the earth a body will have same weight. (take radius of earth as *R*.)

(1) $\frac{\sqrt{5}R - R}{2}$ (2) $\sqrt{5}R - R$

(3) $\frac{R}{2}$ (4) $\frac{\sqrt{3}R - R}{2}$

Answer (1)

Sol. $g\frac{R^2}{(R+h)^2} = g\left(1 - \frac{h}{R}\right)$

$h = \left(\frac{\sqrt{5}-1}{2}\right)R$

34. A biconvex lens of refractive index 1.5 has a focal length of 20 cm in air. Its focal length when immersed in a liquid of refractive index 1.6 will be :

(1) +16 cm (2) +160 cm

(3) –160 cm (4) –16 cm

Answer (3)

Sol. $\frac{1}{f_{air}} = (1.5-1)\left(\frac{2}{R}\right) \Rightarrow R = 20$ cm

$\frac{1}{f_{liq}} = \left(\frac{1.5}{1.6} - 1\right)\left(\frac{2}{20}\right)$

$f_{liq} = -160$ cm

